10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 4 — Human Urinary System

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. The process of regulating the amount of water in body fluid is called:

(a) Filtration
(b) Reabsorption
(c) Tubular secretion
(d) Osmoregulation

Answer: (d) — Osmoregulation means the control of the amount of water according to the needs of the body. Filtration, reabsorption and tubular secretion are the three steps of urine formation.

2. What happens in the proximal convoluted tubule of nephron?

(a) Blood is filtered by glomerulus
(b) Reabsorption of water and nutrients occurs
(c) Urine is stored
(d) Water is secreted into the bloodstream

Answer: (b) — A large amount of water, nearly 100 % of the glucose and amino acids, and most of the salts are reabsorbed from the proximal convoluted tubule into the blood capillaries. Filtration takes place in the glomerulus and urine is stored in the urinary bladder.

3. Which structure is responsible for filtering blood to produce urine?

(a) Nephron
(b) Renal pelvis
(c) Ureter
(d) Renal capsule

Answer: (a) — The nephron is the functional unit of the kidney, and there are more than one million nephrons in each kidney. Its first part, the renal corpuscle, filters the blood; the renal pelvis only collects the urine and the ureter carries it to the bladder.

4. Which structure in nephron is responsible for filtering blood?

(a) Loop of Henle
(b) Glomerulus
(c) Proximal convoluted tubule
(d) Distal convoluted tubule

Answer: (b) — The glomerulus is a network of capillaries inside the cup-shaped Bowman's capsule. The blood pressure in it is very high, so small molecules such as water, salts, glucose and urea are filtered out of it into the Bowman's capsule.

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5. In which part of nephron, the reabsorption of nutrients occurs?

(a) Glomerulus
(b) Proximal convoluted tubule
(c) Loop of Henle
(d) Distal convoluted tubule

Answer: (b) — Nearly 100 % of the glucose and amino acids, a large amount of water and most of the salts are reabsorbed from the proximal convoluted tubule. From the loop of Henle only water and salts are reabsorbed.

6. Which process in nephron involves the active transport of materials from blood into nephron?

(a) Filtration
(b) Reabsorption
(c) Secretion
(d) Excretion

Answer: (c) — In tubular secretion the wastes which are still present in the blood are moved from the blood capillaries into the renal tubule by active transport. Filtration is caused by blood pressure and reabsorption is the return of useful materials to the blood.

7. Which is NOT present in the filtrate entering the Bowman's capsule of nephron?

(a) Water
(b) Calcium ions
(c) Blood cells
(d) Urea

Answer: (c) — Blood cells and proteins are not filtered through the glomerular capillaries because they are relatively larger in size. Water, salts (such as calcium ions), glucose and urea are small molecules, so they pass into the filtrate.

8. In which procedure a machine is used to filter blood outside the body?

(a) Peritoneal dialysis
(b) Haemodialysis
(c) Lithotripsy
(d) Renal surgery

Answer: (b) — In haemodialysis the patient's blood is pumped through a machine called a dialyzer. The blood flows inside long tubes while the dialysis fluid flows outside them, so the wastes and extra water pass into the fluid and the purified blood is returned to the body.

9. What is the primary purpose of haemodialysis?

(a) To rehydrate the body
(b) To replace lost nutrients
(c) To filter waste products from blood
(d) To regulate blood pressure

Answer: (c) — Haemodialysis is used in kidney failure, when the kidneys cannot remove wastes from the blood. The dialyzer filters the waste products and extra water out of the blood, and the clean blood is returned to the body.

10. In peritoneal dialysis, what acts as the filtering membrane?

(a) Dialysis machine (b) Renal cortex

(c) Peritoneal membrane (d) Renal tubules

Answer: (c) — In peritoneal dialysis the dialysis fluid is pumped into the peritoneal cavity, whose walls are lined with the peritoneal membrane. The wastes pass from the blood vessels of this membrane into the dialysis fluid, which is drained after some time.

B. Write short answers.

1. Enlist the main parts of the human urinary system.

The human urinary (excretory) system consists of a **pair of kidneys**, a **pair of ureters**, a **urinary bladder** and a **urethra**. The kidneys remove extra water, salts and nitrogenous wastes from the blood and make urine; each ureter carries urine from a kidney to the urinary bladder, which stores it temporarily, and the urethra carries the urine from the bladder to the outside.

2. Briefly describe how kidneys control the composition of blood.

The kidneys control the chemical composition of the blood by taking the waste materials out of it and converting them into urine. This takes place in three steps. First, pressure filtration occurs in the glomerulus, where water, salts, glucose and urea are filtered into the Bowman's capsule. Then, in reabsorption, the useful materials such as water, glucose, amino acids and most of the salts are taken back into the blood. Finally, in tubular secretion, the remaining wastes are moved from the blood into the renal tubule. In this way only the wastes and the extra substances leave the body in urine, while the useful substances stay in the blood.

3. Name the major parts of a nephron.

A nephron consists of two main parts: the **renal corpuscle** and the **renal tubule**. The renal corpuscle is made of the glomerulus (a network of capillaries) and the Bowman's capsule (a cup-shaped structure around the glomerulus). The renal tubule is a long tube with three parts: the proximal convoluted tubule, the U-shaped loop of Henle and the distal convoluted tubule. The distal convoluted tubules of many nephrons open into a single collecting duct.

4. What materials in diet can cause kidney stones?

Kidney stones are formed when the filtered minerals and salts gather in the kidneys and make hard, crystalline deposits. A diet containing more **calcium oxalate** and more **calcium and ammonium phosphates**—found in green vegetables, fats and dairy products—can cause kidney stones. Extra amounts of **vitamins C and D** in the diet, a reduced intake of water, excess uric acid in the blood and alcohol consumption are also causes of kidney stones.

5. What do you mean by lithotripsy?

Lithotripsy is a method for the removal of kidney stones without surgery. In this method non-electric shock waves are fired on the kidney from outside the body. These waves hit the stones and break them into smaller pieces, which then pass out of the body through the urine. It is used when the stones are not too large; large stones which cannot pass easily have to be removed by surgery.

6. Define kidney failure. Name the treatments for kidney failure.

Kidney failure is the condition in which one or both kidneys are not able to perform their function, that is, they cannot remove the waste materials from the blood and pass them out. As a result the concentration of wastes in the blood increases, the volume of the body fluids rises, and the patient loses weight and has swelling in the legs, feet and face and blood in the urine.

Its treatments are **dialysis** (haemodialysis or peritoneal dialysis) and **kidney transplantation**, in which a healthy kidney from a donor is transplanted into the patient's body.

C. Differentiate between:

i. Excretion and osmoregulation

Excretion	Osmoregulation
It is the removal of waste products and extra substances from the body.	It is the control of the amount of water in the body fluids, according to the needs of the body.
The wastes removed are urea, extra salts, carbon dioxide and water.	Only water and salts are controlled, so that their concentration stays normal.
It is carried out by the kidneys, lungs, skin and alimentary canal.	It is carried out mainly by the kidneys.
Its result is urine, which carries the wastes out of the body.	Its result may be dilute or concentrated urine, depending on whether the body needs to remove or to keep water.

ii. Renal cortex and renal medulla

Renal cortex	Renal medulla
It is the outer region of the kidney.	It is the inner region of the kidney.
It contains the renal corpuscles, that is, the glomerulus and the Bowman's capsule, along with the convoluted tubules.	It contains the cone-shaped renal pyramids and the loops of Henle, along with the collecting ducts.
Filtration of the blood takes place here.	Concentration of the urine takes place here, because most of the water is reabsorbed from this region.

iii. Renal corpuscle and renal tubule

Renal corpuscle	Renal tubule
It is the first part of a nephron.	It is the second part of a nephron, attached to the corpuscle.
It is made of the glomerulus and the Bowman's capsule.	It is made of the proximal convoluted tubule, the loop of Henle and the distal convoluted tubule.
Pressure filtration takes place here; blood is filtered into the Bowman's capsule.	Reabsorption and tubular secretion take place here; useful materials go back into the blood and wastes are added to the filtrate.

iv. Pressure filtration and tubular secretion

Pressure filtration	Tubular secretion
It is the first step of urine formation and takes place in the glomerulus.	It is the last step of urine formation and takes place in the renal tubule.
Materials move out of the blood because of the high blood pressure in the glomerulus.	Materials move from the blood capillaries into the renal tubule by active transport.
Water, salts, glucose and urea are filtered into the Bowman's capsule.	The wastes still present in the blood, such as excess salts, drugs and hydrogen ions, are added to the filtrate.
No energy is spent directly; the pressure of the blood does the work.	Energy is spent, because it is an active process.

v. Haemodialysis and peritoneal dialysis

Haemodialysis	Peritoneal dialysis
The patient's blood is pumped through a machine called a dialyzer.	The dialysis fluid is pumped into the peritoneal cavity of the patient.
Blood flows inside long tubes of the dialyzer and the dialysis fluid flows outside them.	The peritoneal membrane, which lines the peritoneal cavity, acts as the filtering membrane.
The wastes and extra water move from the blood into the dialysis fluid, and the purified blood is returned to the body.	The wastes pass from the blood vessels of the peritoneal membrane into the dialysis fluid, which is drained out after some time.
It is done in a hospital or dialysis centre, with the help of a machine.	It can often be done at home, and no machine is needed for filtering the blood.

vi. Label the parts of the given diagrams (A) and (B).

The two diagrams show the human urinary system and the internal structure of a kidney. The parts to be labelled are:

Diagram A — Urinary system	Diagram B — Section of kidney
Kidney	Renal capsule
Ureter	Renal cortex
Urinary bladder	Renal medulla
Urethra	Renal pyramids
Renal artery and renal vein	Renal pelvis and ureter

D. Write answers in detail.

1. Write a note on the structure of kidney.

Position and size: kidneys are the major organs of the urinary system. They are bean-shaped organs with a concave and a convex side, present in the abdominal cavity below the diaphragm, on the sides of the vertebral column. Their concave side faces the vertebral column, and the left kidney is a little higher than the right. Kidneys are dark red in colour, and each kidney is about 10 cm long, 5 cm wide and 4 cm thick.

External structure: each kidney is surrounded by a tough tissue called the renal capsule. On its concave side there is a depression called the hilus, and it is the point from where the renal artery enters the kidney and the renal vein and the ureter leave it.

Internal structure: internally the kidney is divided into two regions: the outer region is the renal cortex and the inner one is the renal medulla. There are many cone-shaped areas in the renal medulla called renal pyramids; the base of each pyramid faces the renal cortex, while its tip lies in a cavity called the renal pelvis. The renal pelvis extends to the outside of the kidney and forms the ureter.

Nephrons: the functional units of the kidney are called nephrons, and there are more than one million nephrons in each kidney. Each nephron consists of a renal corpuscle (glomerulus and Bowman's capsule) and a renal tubule. Blood enters the kidney through the renal artery and is filtered in the nephrons, and the urine formed is collected by the collecting ducts into the renal pelvis and then carried by the ureter to the bladder.

2. Describe the structure of a nephron and draw a labelled diagram.

The nephron is the functional unit of the kidney. Each kidney contains more than one million nephrons, and each nephron has two main parts: the renal corpuscle and the renal tubule.

1. Renal corpuscle: it is the first part of the nephron and is composed of the glomerulus and the Bowman's capsule. The glomerulus is a network of capillaries, and the Bowman's capsule is the cup-shaped structure around the glomerulus. Blood reaches the glomerulus through an arteriole, and the pressure of the blood in it is

very high.

2. Renal tubule: it is a long tube attached to the Bowman's capsule and has three parts. The first part is convoluted and is called the proximal convoluted tubule; the middle part is U-shaped and is called the loop of Henle; and the last part, which is again convoluted, is called the distal convoluted tubule. The distal convoluted tubules of many nephrons open into a single collecting duct, and many collecting ducts join together and open into the renal pelvis.

Labelled diagram: the diagram should show the arteriole entering the glomerulus inside the Bowman's capsule, then the proximal convoluted tubule, the loop of Henle (descending and ascending limbs), the distal convoluted tubule, the collecting duct, and the capillaries surrounding the tubule that lead to the venule.

3. Describe the process of pressure filtration and re-absorption in kidneys.

Pressure filtration: blood enters the kidney through the renal artery, and the artery divides into many arterioles. Each arteriole divides into the capillaries of the glomerulus. When the blood reaches the glomerulus, its pressure is very high, so many small molecules such as water, salts, glucose and urea move out of the glomerulus and enter the Bowman's capsule. This step is called pressure filtration, and the filtered material is called the glomerular filtrate. It is about 20 % of the blood plasma that enters the glomerulus. Blood cells and proteins are not filtered, because they are relatively larger in size.

Re-absorption: the glomerular filtrate now moves into the renal tubule, where many useful materials are reabsorbed from it into the blood capillaries surrounding the tubule. A large amount of water, nearly 100 % of the glucose and amino acids and most of the salts are reabsorbed from the proximal convoluted tubule. Some water is reabsorbed from the descending limb of the loop of Henle and large amounts of salts from its ascending limb, and some water is also reabsorbed from the distal convoluted tubule and the collecting duct. In this way the useful materials do not leave the body, and only the wastes go out in the urine.

4. What is tubular secretion? How it helps in removing wastes from blood?

After the reabsorption of useful materials into the blood, some of the wastes still present in the blood are moved from the blood capillaries into the renal tubule. This process is called tubular secretion, and it takes place by active transport, so energy is spent in it. The substances secreted include the wastes which were not filtered in the glomerulus, such as excess salts, hydrogen ions and some drugs.

Tubular secretion helps in removing wastes from the blood because it gives a second chance for the removal of those wastes which escaped filtration. It also helps to control the pH and the salt composition of the blood. After reabsorption and tubular secretion, the filtrate present in the renal tubules is called urine, which moves to the renal pelvis and then through the ureter to the urinary bladder, from where it is passed out through the urethra.

5. Write a comprehensive note on kidney stones describing the causes, symptoms and treatments.

What they are: sometimes the filtered minerals and salts, such as calcium oxalate, calcium and ammonium phosphate and uric acid, gather in the kidneys and make hard, crystalline deposits. Such deposits cannot pass in the urine and are called kidney stones. Some stones may leave the kidney and be trapped in the ureter or in the urinary bladder.

Symptoms: severe pain in the kidney or lower abdomen, burning in the urethra, frequent urination, foul-smelling urine, blood and pus in the urine, nausea, vomiting and bloating.

Causes: a diet with more calcium oxalate and more calcium and ammonium phosphates (green vegetables, fats and dairy products), extra amounts of vitamins C and D, a reduced intake of water, excess uric acid in the blood, urinary tract infections and alcohol consumption.

Treatment: if the stones are small, the patient is advised to drink plenty of water so that they pass out in the urine. Lithotripsy is used for medium-sized stones: non-electric shock waves are fired on the kidney from outside and they break the stones into smaller pieces which pass out through the urine. If the stones are large and cannot pass easily, the patient has to undergo surgery for their removal.

6. Write down the symptoms and causes of kidney failure.

Symptoms: an increased concentration of wastes in the blood, a rise in the volume of the body fluids, weight loss, blood in the urine, and swelling in the legs, feet and face.

Causes: long-term diabetes mellitus and hypertension (high blood pressure), dehydration, severe kidney infections, overuse of certain medicines such as nonsteroidal anti-inflammatory drugs (NSAIDs), large kidney stones, physical damage to the kidneys, and heart or liver failure.

7. Define dialysis and describe the processes of peritoneal dialysis and haemodialysis.

Dialysis is a method in which the wastes are extracted from the blood by using a dialysis fluid. It is used when the kidneys have failed and cannot remove the wastes themselves. It has two methods.

Haemodialysis: in this method the patient's blood is pumped through a machine called a dialyzer. The dialyzer contains long tubes; the blood flows inside the tubes while the dialysis fluid flows outside them. Because the concentration of wastes is high in the blood and zero in the dialysis fluid, the extra water and wastes move from the blood into the dialysis fluid. The purified blood is then returned to the body.

Peritoneal dialysis: in this method the dialysis fluid is pumped into the peritoneal cavity, which is the space around the gut in the abdomen. The walls of this cavity are lined with the peritoneal membrane, and the wastes pass from the blood in the blood vessels of this membrane into the dialysis fluid. After some time the used dialysis fluid is drained out and fresh fluid is put in.

E. Inquisitive questions

1. “Along with excretion, kidneys also play role in osmoregulation.” Comment on this statement.

The statement is correct. Kidneys remove the nitrogenous wastes such as urea, along with extra water and salts, in the form of urine; this is their excretory function. At the same time they control the amount of water and salts in the body fluids, and this is called osmoregulation. When the body has a lot of water, the kidneys filter more water and produce dilute, large amounts of urine. When the body is short of water, less water is filtered from the glomerulus and more water is reabsorbed from the renal tubules, so concentrated urine is produced and the water is retained in the blood. In this way the kidneys keep the internal environment of the body stable.

2. How does the nephron ensure that essential nutrients are not lost in urine?

The nephron works in stages, and each stage saves the useful materials. In the glomerulus, only the small molecules are filtered, so blood cells and the large protein molecules stay in the blood. In the proximal convoluted tubule, nearly 100 % of the glucose and amino acids, a large amount of water and most of the salts are reabsorbed into the blood capillaries. Some water is reabsorbed from the descending limb of the loop of Henle and salts from the ascending limb, and more water is reabsorbed from the distal convoluted tubule and the collecting duct. In this way, by the time the filtrate becomes urine, almost all the useful substances have gone back into the blood, and only the wastes and excess substances are left to be excreted.

3. Why is long loop of Henle important in the balance of water and salts in body?

The loop of Henle is U-shaped and has two limbs. In the descending limb, water is reabsorbed from the filtrate, and in the ascending limb large amounts of salts are reabsorbed. Because of these two opposite movements, the fluid in the medulla of the kidney becomes concentrated, and this concentration helps the collecting duct to reabsorb more water from the filtrate. A long loop of Henle therefore enables the kidney to reabsorb a large amount of water and to produce concentrated urine, which saves water for the body. In this way the loop of Henle keeps the balance of both water and salts in the body.

4. How can chronic diseases like diabetes lead to kidney failure?

In diabetes mellitus the concentration of glucose in the blood remains high for a long time. The kidneys filter this excess glucose, and much of it cannot be reabsorbed, so it passes out in the urine. This makes the urine very concentrated and increases the amount of urine, which causes dehydration. The excess glucose also damages the small blood vessels of the glomeruli, so the filters of the kidney become weak and leak protein and blood into the urine. Because of this long-term damage the kidney cannot filter the blood properly, waste products collect in the blood and, if it is not controlled, the condition progresses to kidney failure. That is why diabetic patients are advised to keep their blood sugar under control.

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